9/08

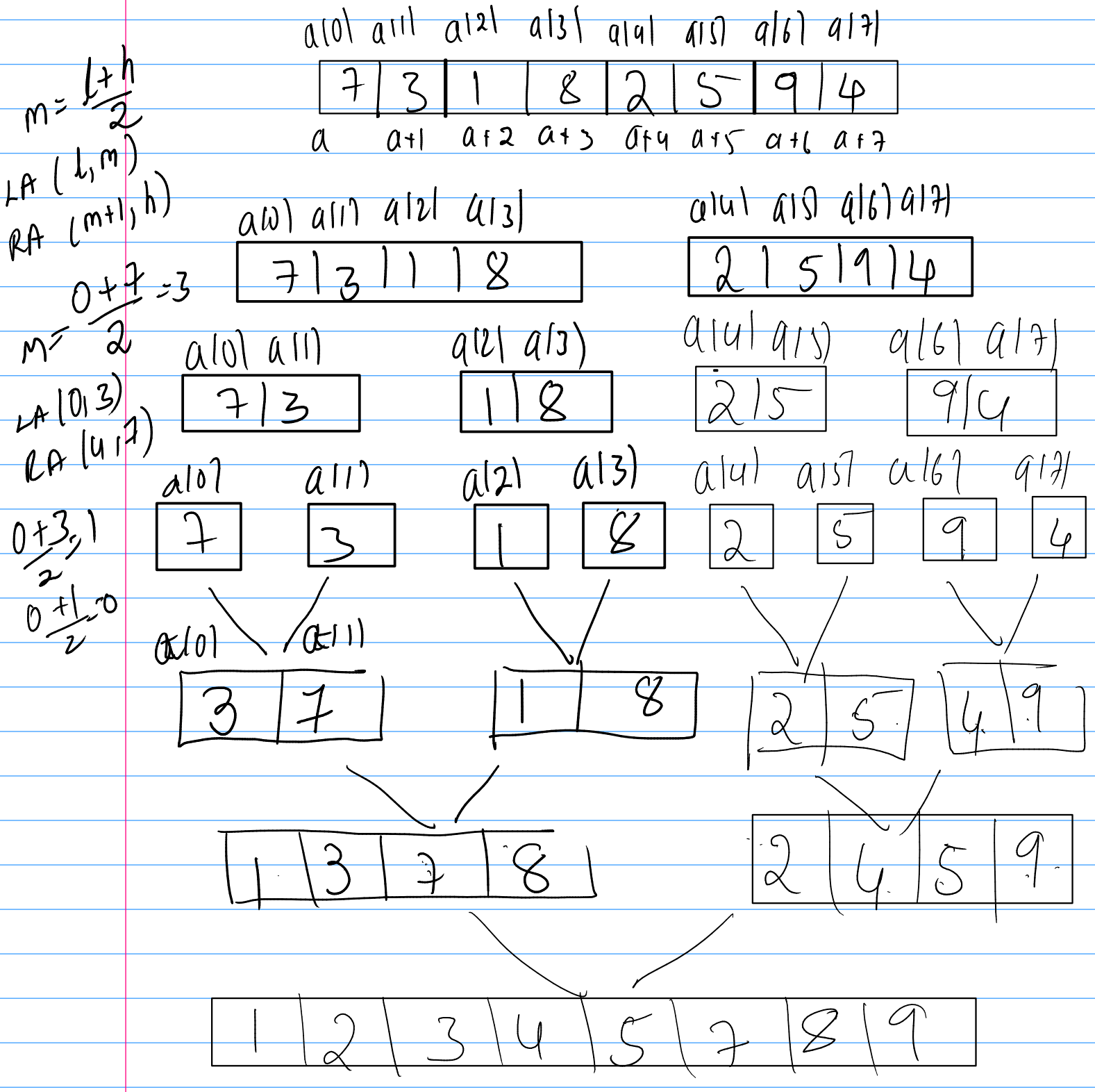
Merge Sort

Merge Sort follows divide and conquer method

Larger array will broken into smaller arrays.

How much smaller until we get the array with single element

now merge the consecutive sorted arrays

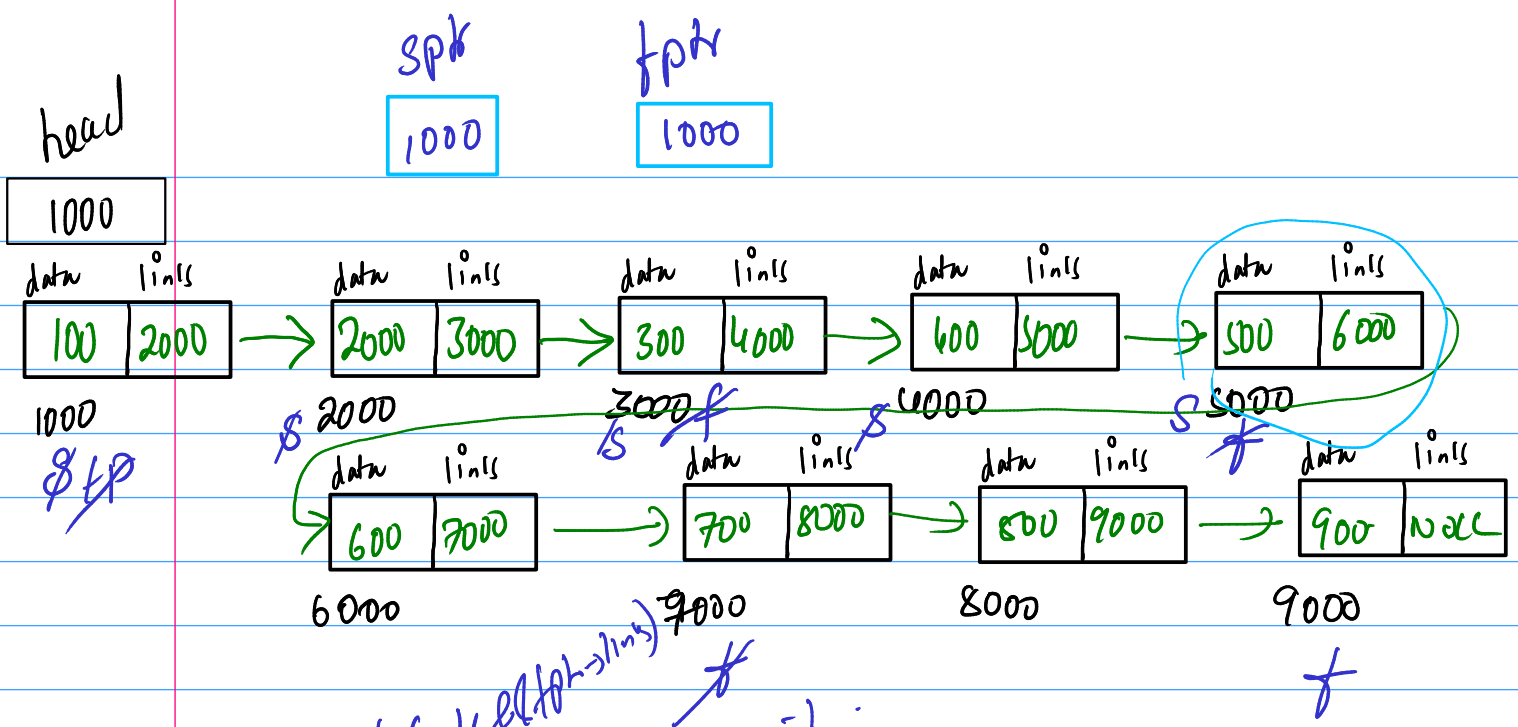


1. Divide the array into equal parts until we get the array with single element.
2. merge the consecutive sorted arrays into temporay array
3. update the sorted elements from temporay array to actual array

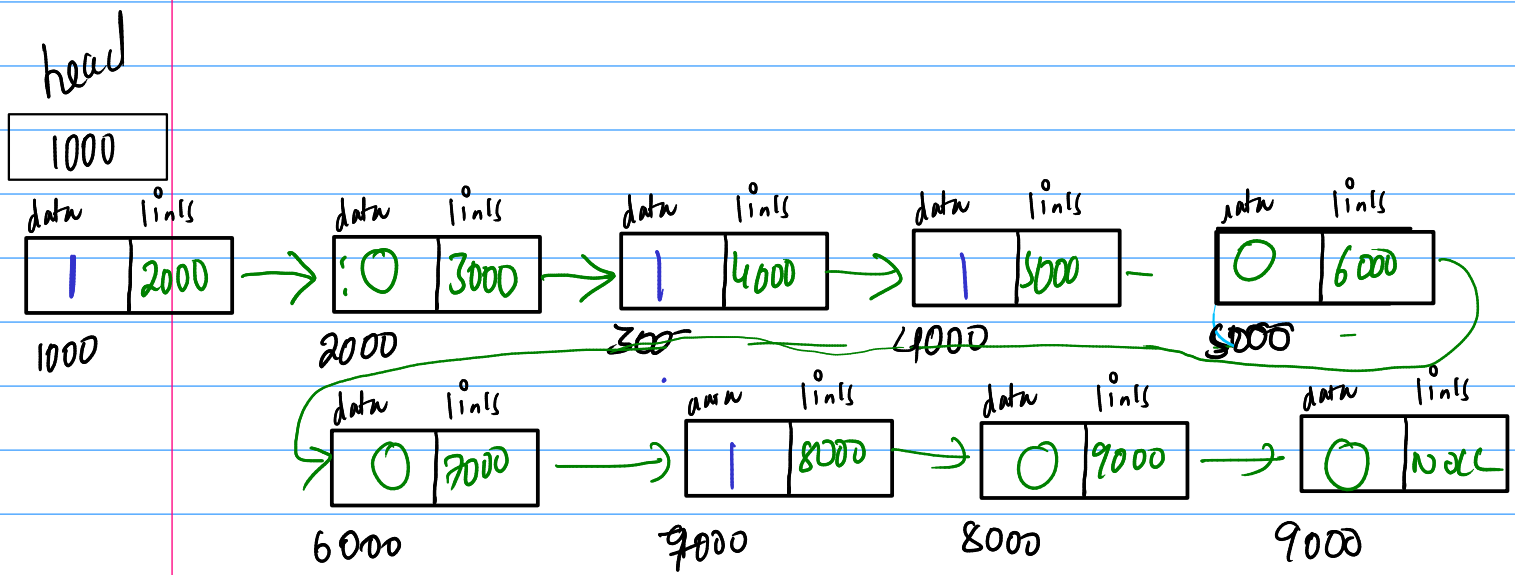
```
dividing of array
if(low!=high)
m=(low+high)/2;
partition(a,low,m);
partition(a,m+1,high);
merge(a,low,m,high);

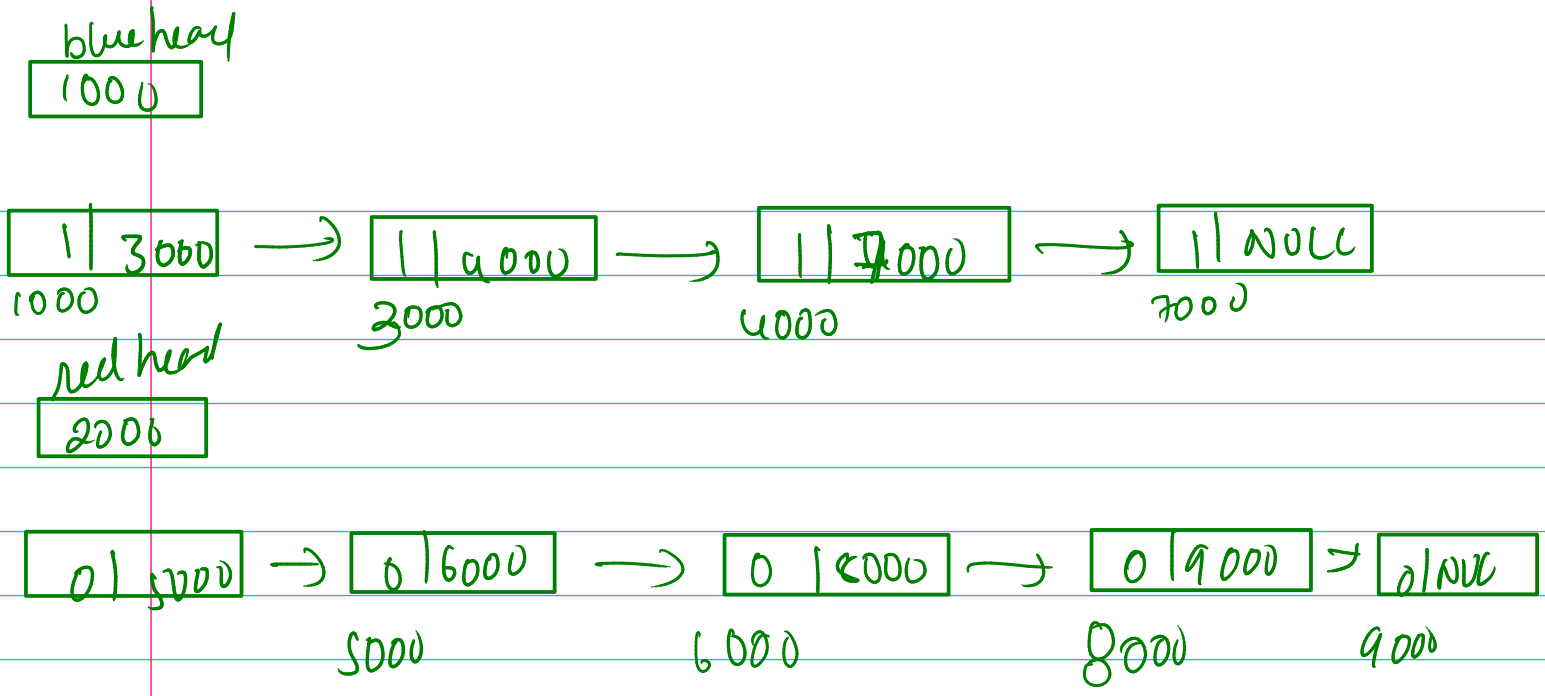
while(li<=m && ri<=high)
{           if(a[li] < a[ri])
            {
                        temp[i]=a[li];
                        li++;
            }
            esle
            {           temp[i]=a[ri];
                        ri++;
            }
            i++;
}

if(li>m)
append elements of right arr
else
append the elements of left arr
update the elemenst from temp
array to actual array
```




```
while (fpt && (fpt->link))
{ p = spt;
  spt = spt->link;
  fpt = fpt->link->link;
}
if (fpt == NULL)
  printf("%d", p->data);
else
  printf("%d", spt->data);
```




blue head
1000
1 | 3000
1000
1 | 4000
3000
1 | 7000
4000
1 | NULL
7000
red head
2000
0 | 5000
0 | 6000
5000
0 | 8000
6000
0 | 9000
8000
0 | NULL
9000